

SECTION 9

OPERATIONAL INFORMATION

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SECTION 9

OPERATIONAL INFORMATION

9.1 GENERAL

This section deals with additional information to supplement Section 5 (Performance data), provide assistance in preflight planning, and be helpful in estimating the remaining flight time or range if, for some reason, the originally intended flight plan must be changed during flight.

For definition of terms, abbreviations and symbols used in this section, refer to Section 1.

9.1.1 Standard Performance Conditions

The data presented cover the maximum range of performance that can be expected and are based on the following conditions:

- Engine power does not exceed limits (see Section 2).
- Installation and accessory losses are included
- Rotor RPM - 100%

In each area of performance, the effects of altitude, temperature, gross mass, and other parameters relating to that phase of flight are presented. In addition to the presented data, judgment and experience will be necessary to accurately determine performance under a given set of circumstances.

9.1.2 Conditions Affecting Performance

Conditions affecting performance are listed in the heading of the respective charts. The data presented are accurate only for these specific conditions.

Other general conditions applicable to the performance data are:

- Airframe and engine control rigging (assumed within allowable tolerances)
- Pilot technique (assumed normal, control movements smooth and continuous)
- Variations in performance between individual helicopters
- Instrument variation (inaccuracies or malfunctions are not accounted for)

9.1.3 Reading the Charts

None of the curves presented should be extrapolated, however, interpolation between given data is permissible.

9.2 FUEL CONSUMPTION

The fuel consumption diagrams (see Fig. 9-1 thru 9-9) are based on the following conditions:

- Fuel density is 0.8 kg/l (6.682 lb/U.S. gal.)
- The fuel used meets one of the approved fuel types as specified in Section 2.

FUEL CONSUMPTION

HOVER IN GROUND EFFECT

2 X ALLISON 250-C20B

ANTI-ICING - OFF

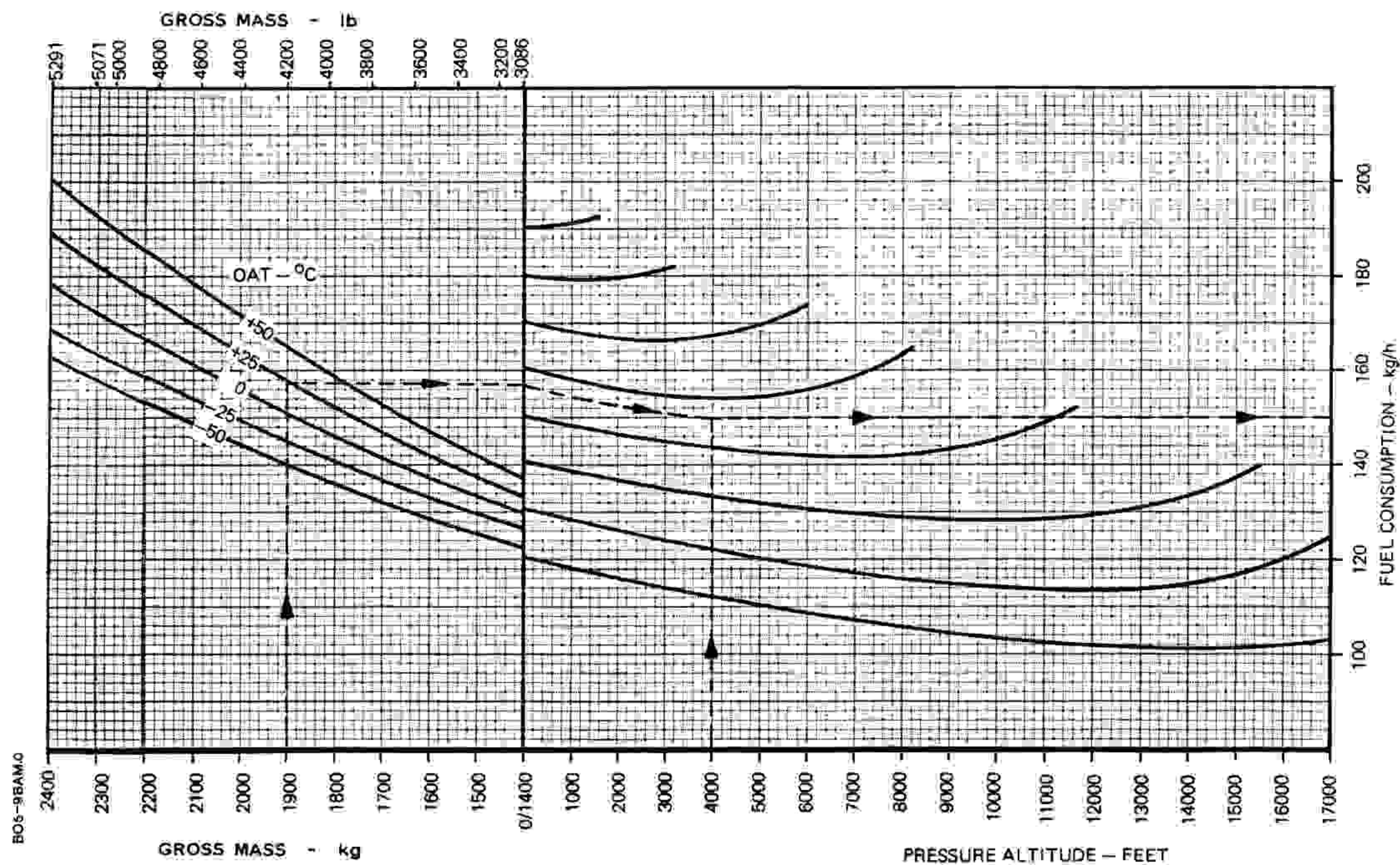


Fig. 9-1 Fuel Consumption - H I G E

FUEL CONSUMPTION

CRUISE FLIGHT, 0-80 KIAS

2 X ALLISON 250-C20B

OAT RANGE: 0 TO +50 °C

ANTI-ICING — OFF

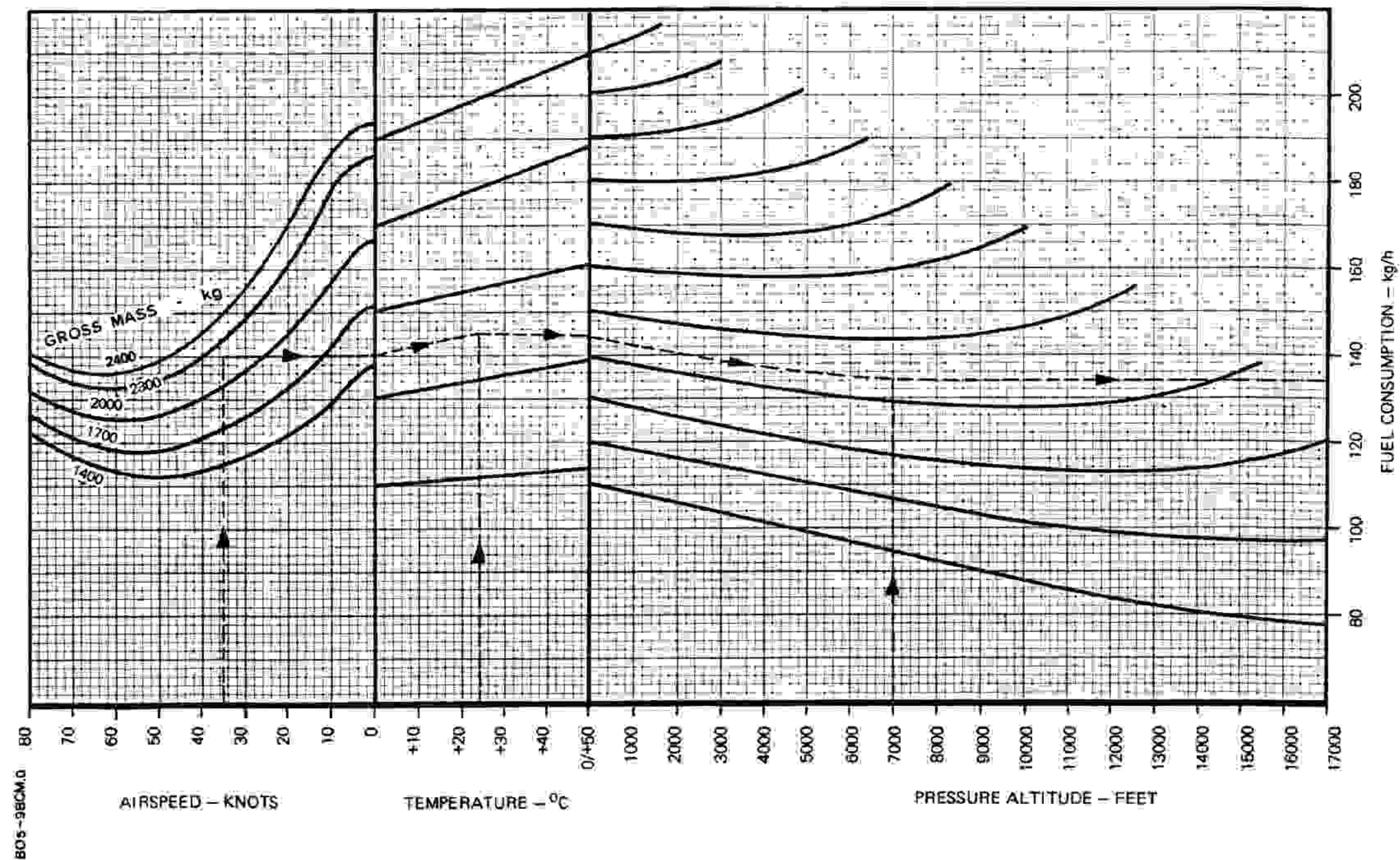


Fig. 9-2 Fuel Consumption - Cruise Flight, 0-80 KIAS, 0 to +50 °C

FUEL CONSUMPTION

CRUISE FLIGHT, 0-80 KIAS

2 X ALLISON 250-C20B

OAT RANGE: -50 TO 0 °C

ANTI-ICING - OFF

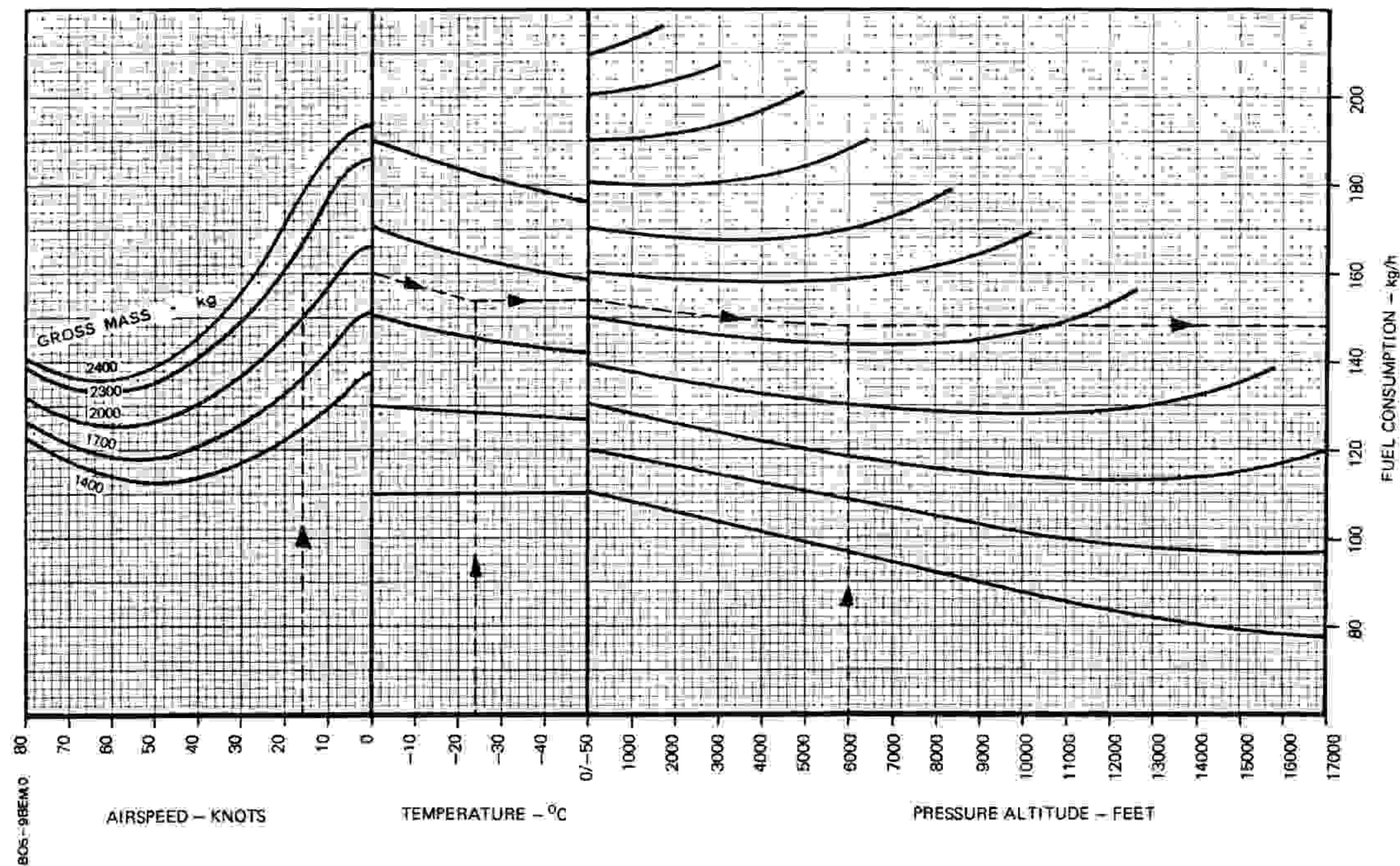


Fig. 9-3 Fuel Consumption - Cruise Flight, 0-80 KIAS, -50 to 0 °C

FUEL CONSUMPTION

CRUISE FLIGHT, ABOVE 70 KIAS

2 X ALLISON 250-C20B

OAT RANGE: -50 TO +10°C

ANTI-ICING - ON

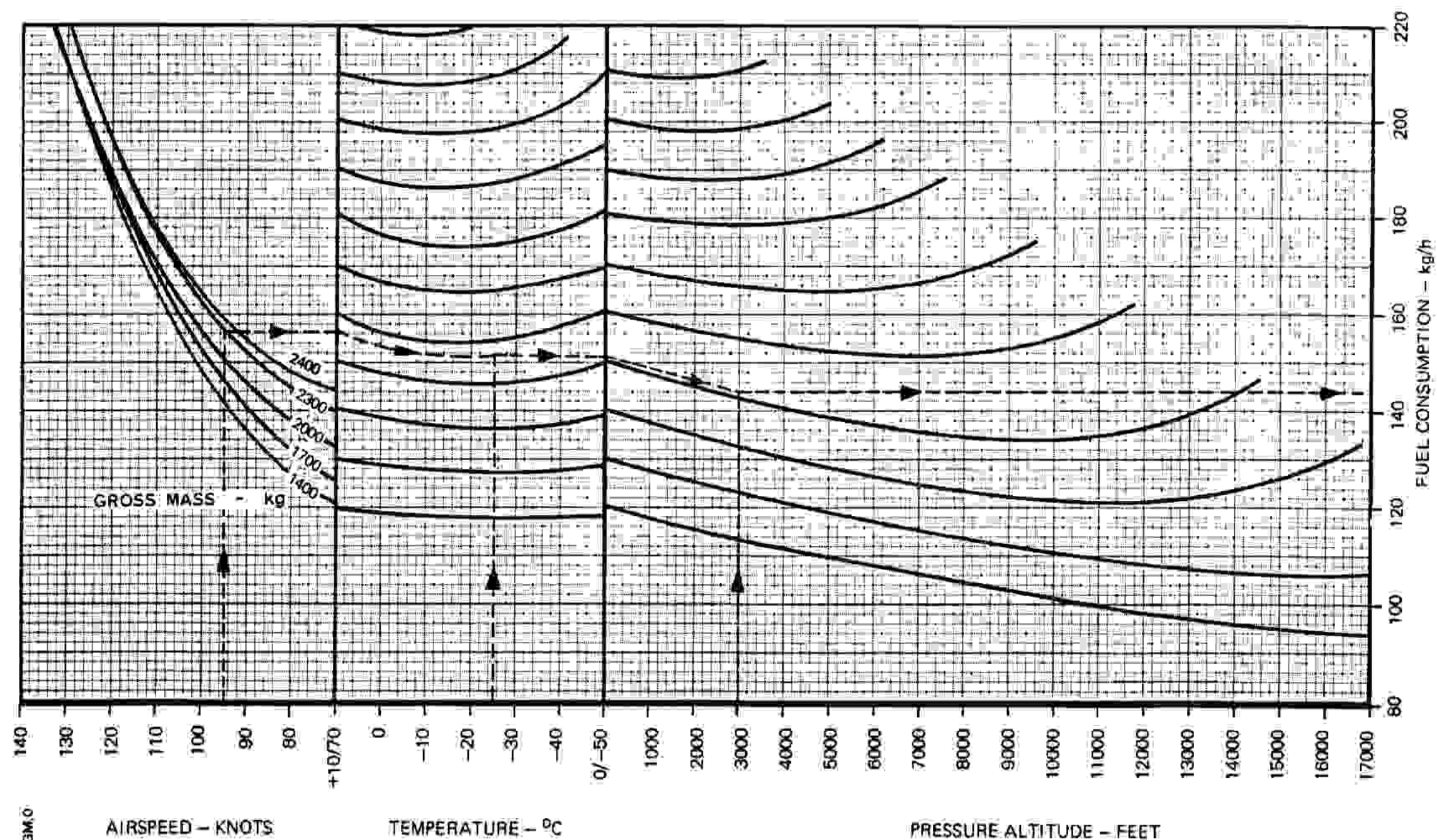


Fig. 9-4 Fuel Consumption - Cruise Flight, Above 70 KIAS, Anti-icing ON

FUEL CONSUMPTION

CRUISE FLIGHT, ABOVE 70 KIAS

2 X ALLISON 250-C20B

OAT RANGE: 0 TO +50 °C

ANTI-ICING - OFF

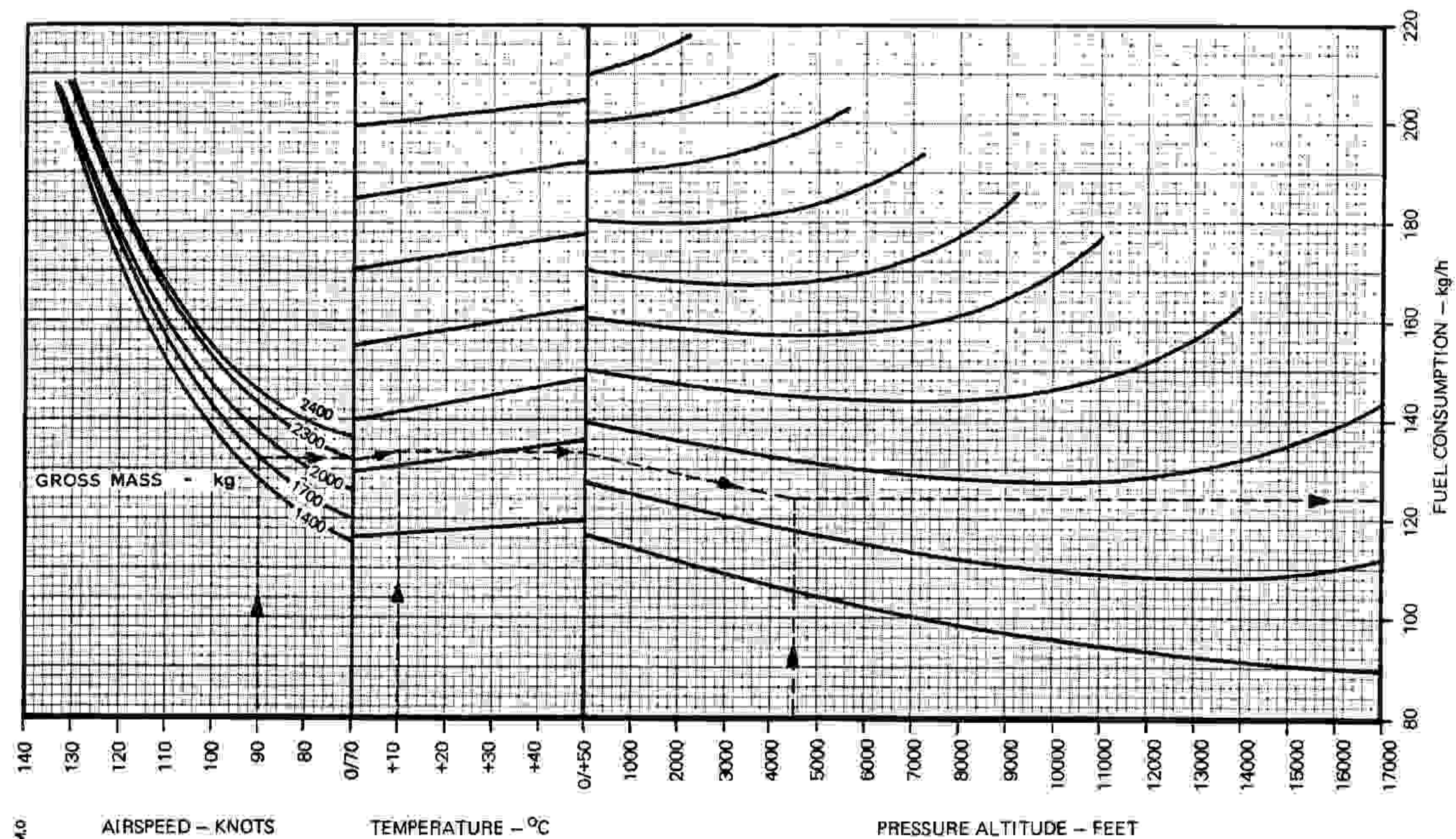


Fig. 9-5 Fuel Consumption - Cruise Flight, Above 70 KIAS, 0 to +50 °C

FUEL CONSUMPTION

CRUISE FLIGHT, ABOVE 70 KIAS

2 X ALLISON 250-C20B

OAT RANGE: -50 TO 0 °C

ANTI-ICING - OFF

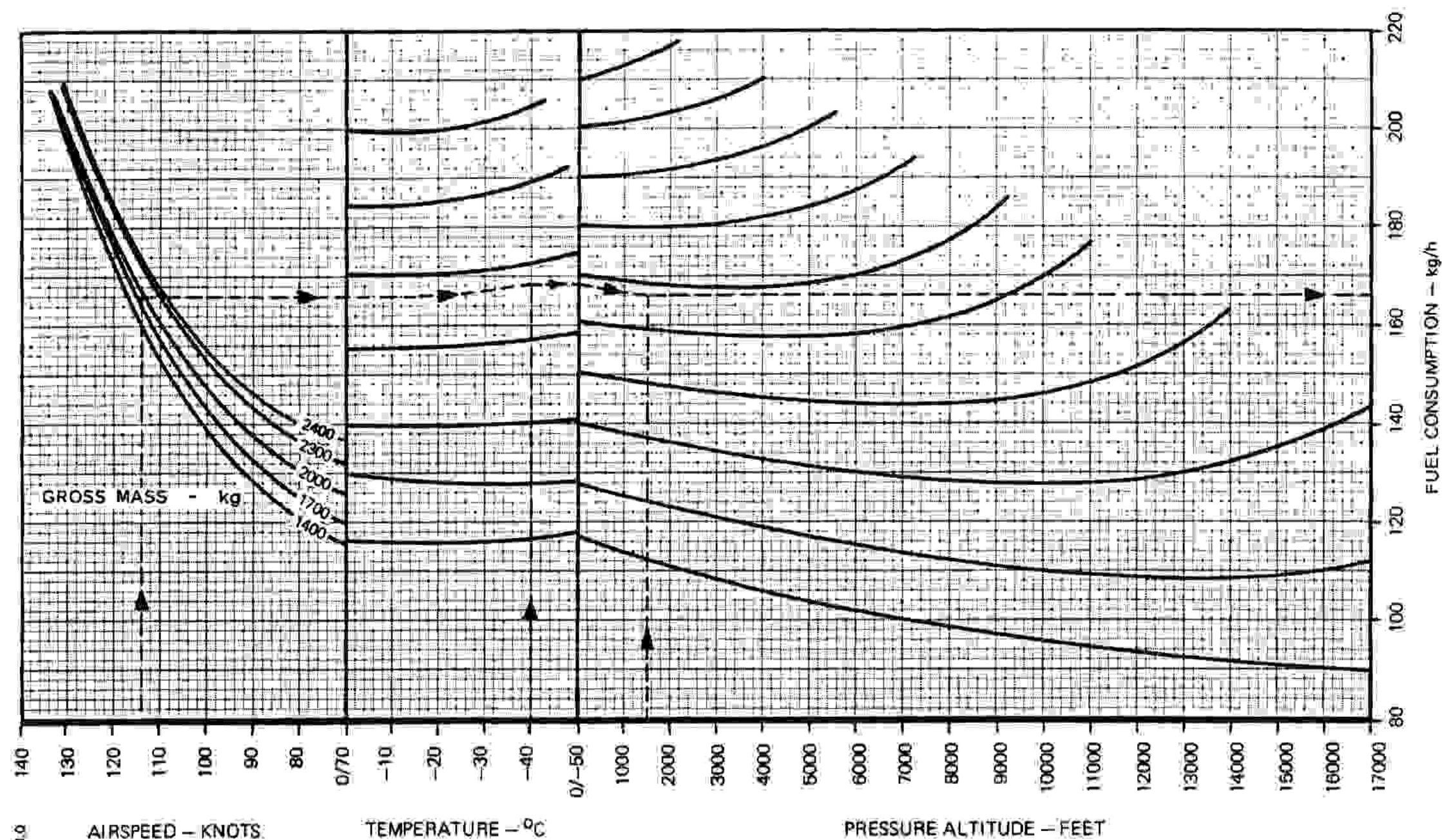


Fig. 9-6 Fuel Consumption - Cruise Flight, Above 70 KIAS, -50 to 0 °C

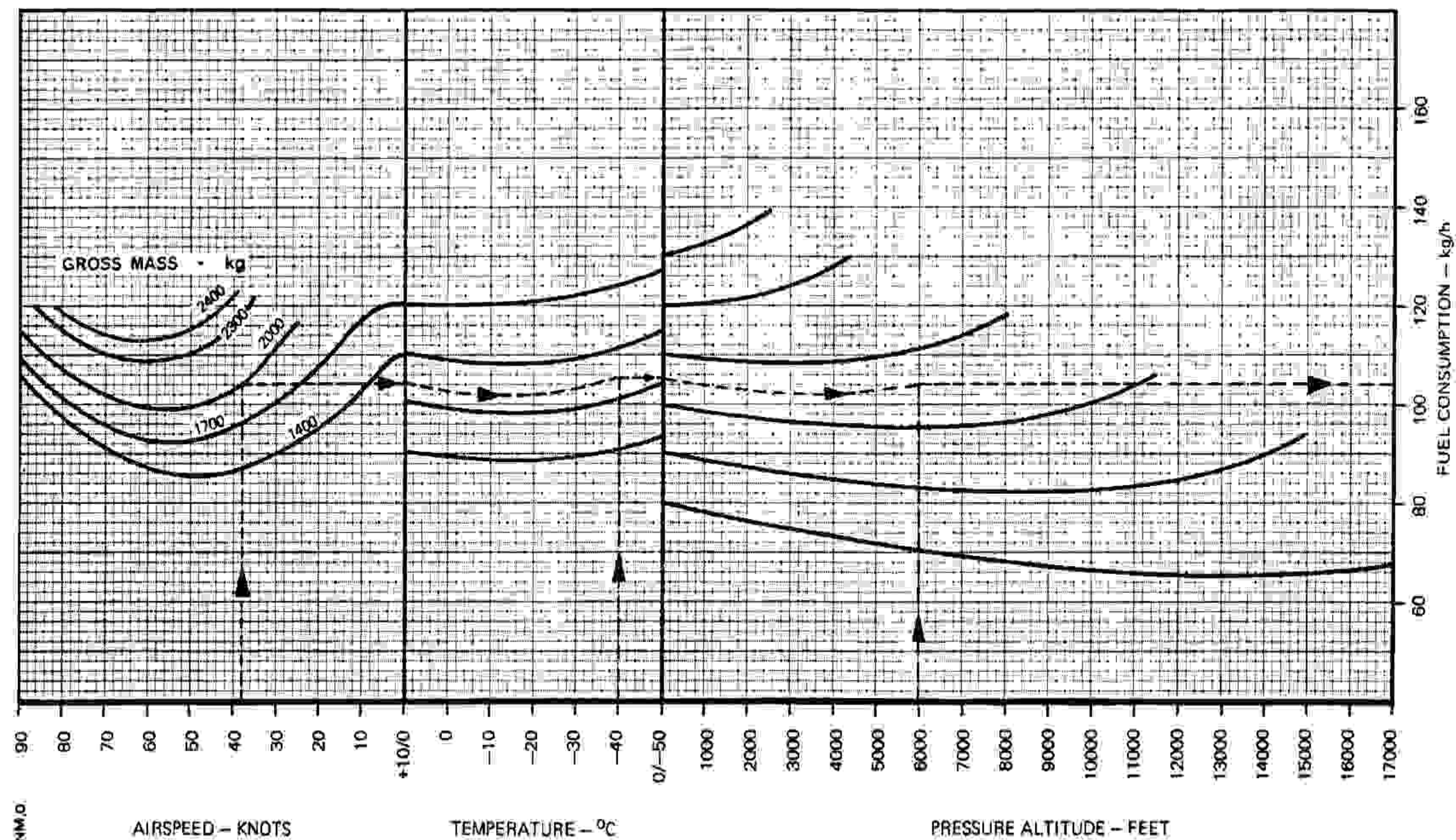
OEI FUEL CONSUMPTION

CRUISE FLIGHT, 0-90 KIAS

1 X ALLISON 250-C20B

OAT RANGE: -50 TO +10 °C

ANTI-ICING - ON



BO5-98NM.0

Fig. 9-7 OEI Fuel Consumption - Cruise Flight, 0-90 KIAS, Anti-icing ON

MANUFACTURER'S DATA

Rev. 0

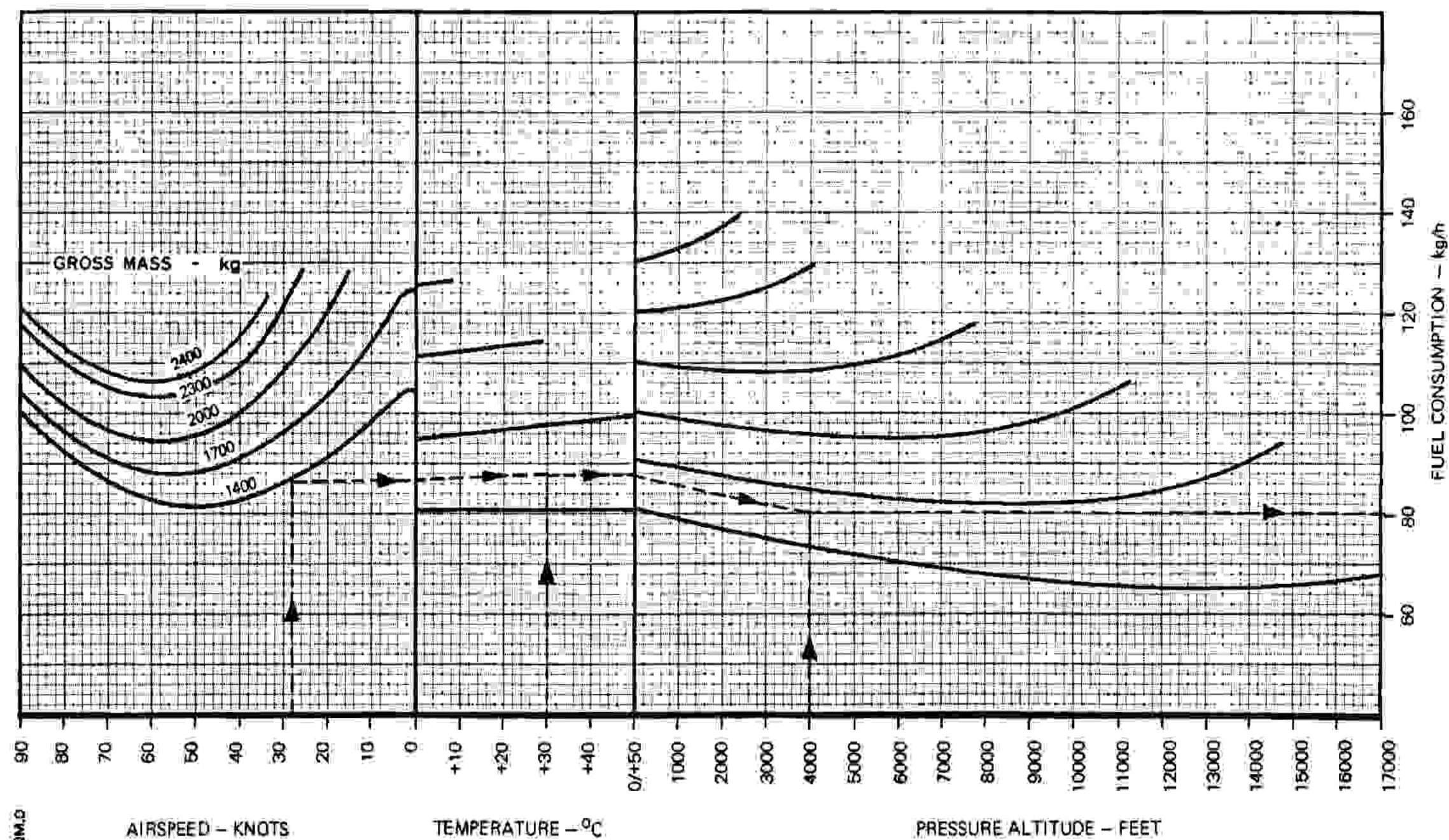
OEI FUEL CONSUMPTION

CRUISE FLIGHT, 0-90 KIAS

1 X ALLISON 250-C20B

OAT RANGE: 0 TO +50 °C

ANTI-ICING - OFF



BOS-BBQM.0

Fig. 9-8 OEI Fuel Consumption - Cruise Flight, 0-90 KIAS, 0 to +50 °C

OEI FUEL CONSUMPTION

CRUISE FLIGHT, 0-90 KIAS

1 X ALLISON 250-C20B

OAT RANGE: -50 TO 0 °C

ANTI-ICING - OFF

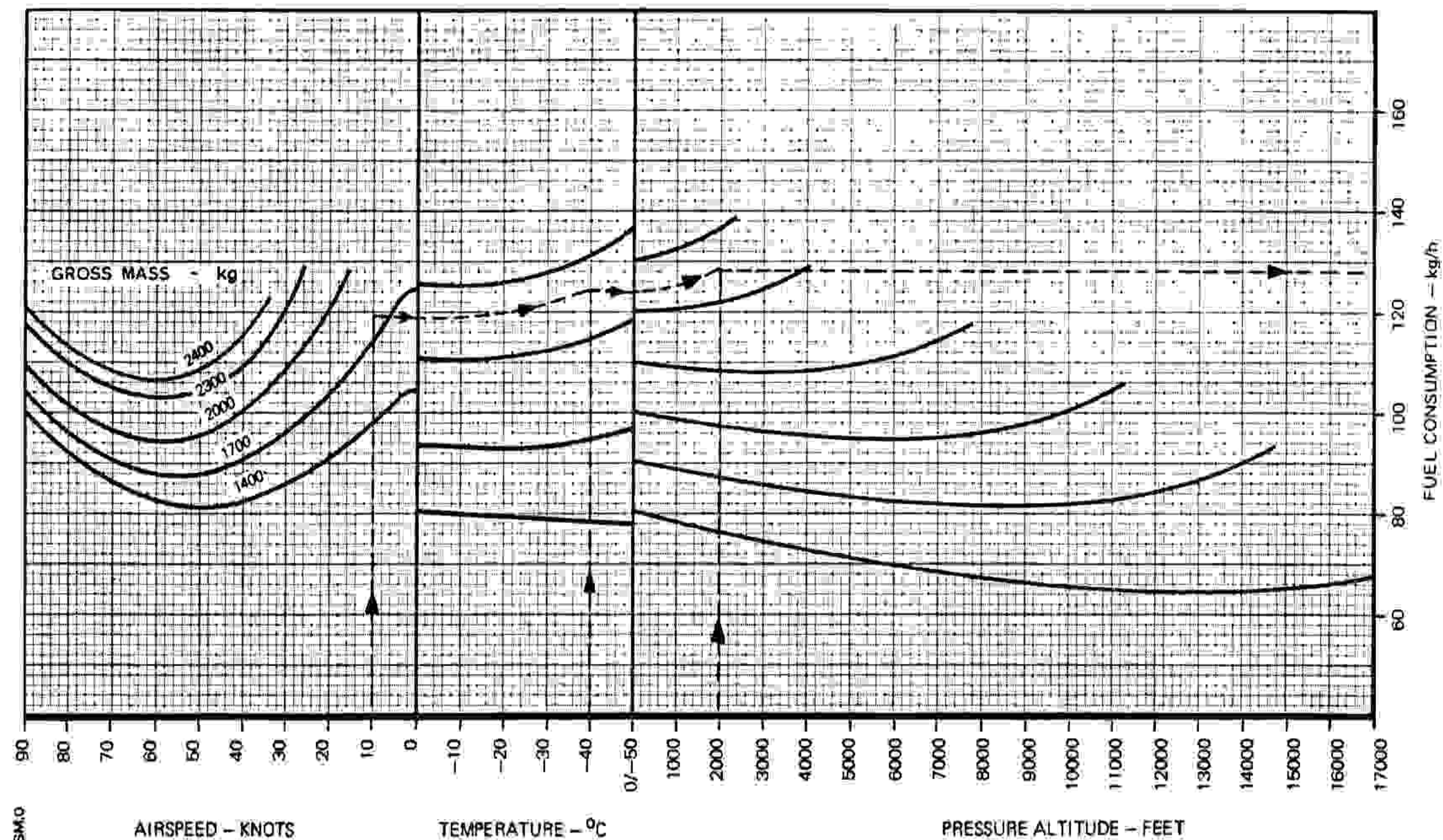


Fig. 9-9 OEI Fuel Consumption - Cruise Flight, 0-90 KIAS, -50 to 0 °C

9.3 MAXIMUM RANGE

The maximum range diagrams (see Fig. 9-10 and 9-11) are based on the following conditions:

- International Standard Atmosphere (ISA)
- Usable fuel ~ 456 kg (no reserve)
- Level flight (no climb, descent, or other maneuver)
- Engine anti-icing - OFF

NOTE With anti-icing ON, the maximum range will be reduced.

For range calculations at altitudes and fuel loads other than those presented in the diagrams, determine true airspeed ($IAS \times \frac{1}{\sigma}$), fuel consumption and present fuel quantity.

Calculate range using the formula:

$$\text{Range} = \frac{\text{TAS} \times \text{Fuel Quantity}}{\text{Fuel Consumption}}$$

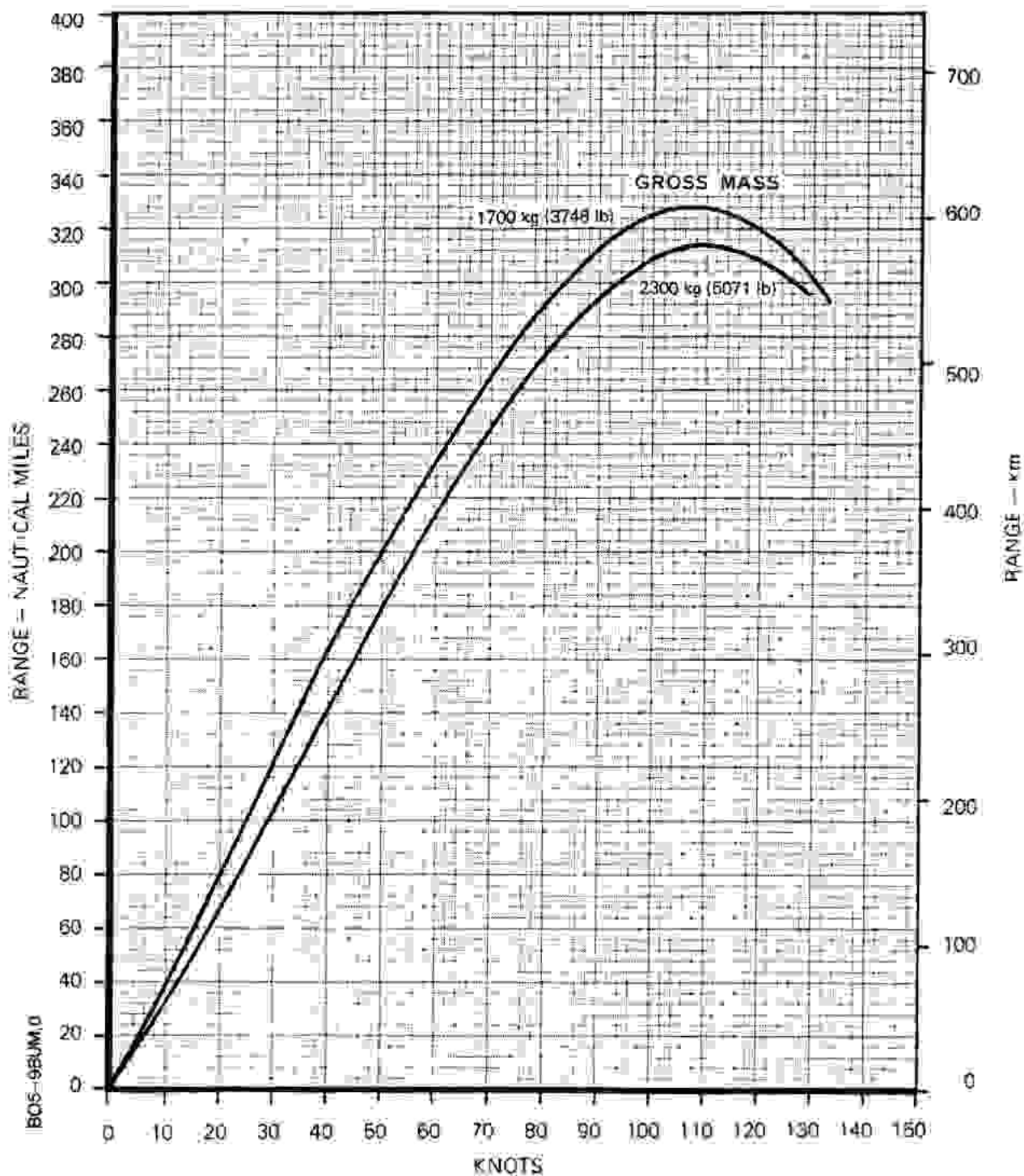
MAXIMUM RANGE**PRESSURE ALTITUDE - ISA SEA LEVEL****2 X ALLISON 250-C20B****FUEL - 456 KG; NO RESERVE****ANTI-ICING - OFF**

Fig. 9-10: Maximum Range - ISA Sea Level

MAXIMUM RANGE

PRESSURE ALTITUDE - ISA 5000 FT

2 X ALLISON 250-C20B

FUEL - 456 KG, NO RESERVE

ANTI-ICING - OFF

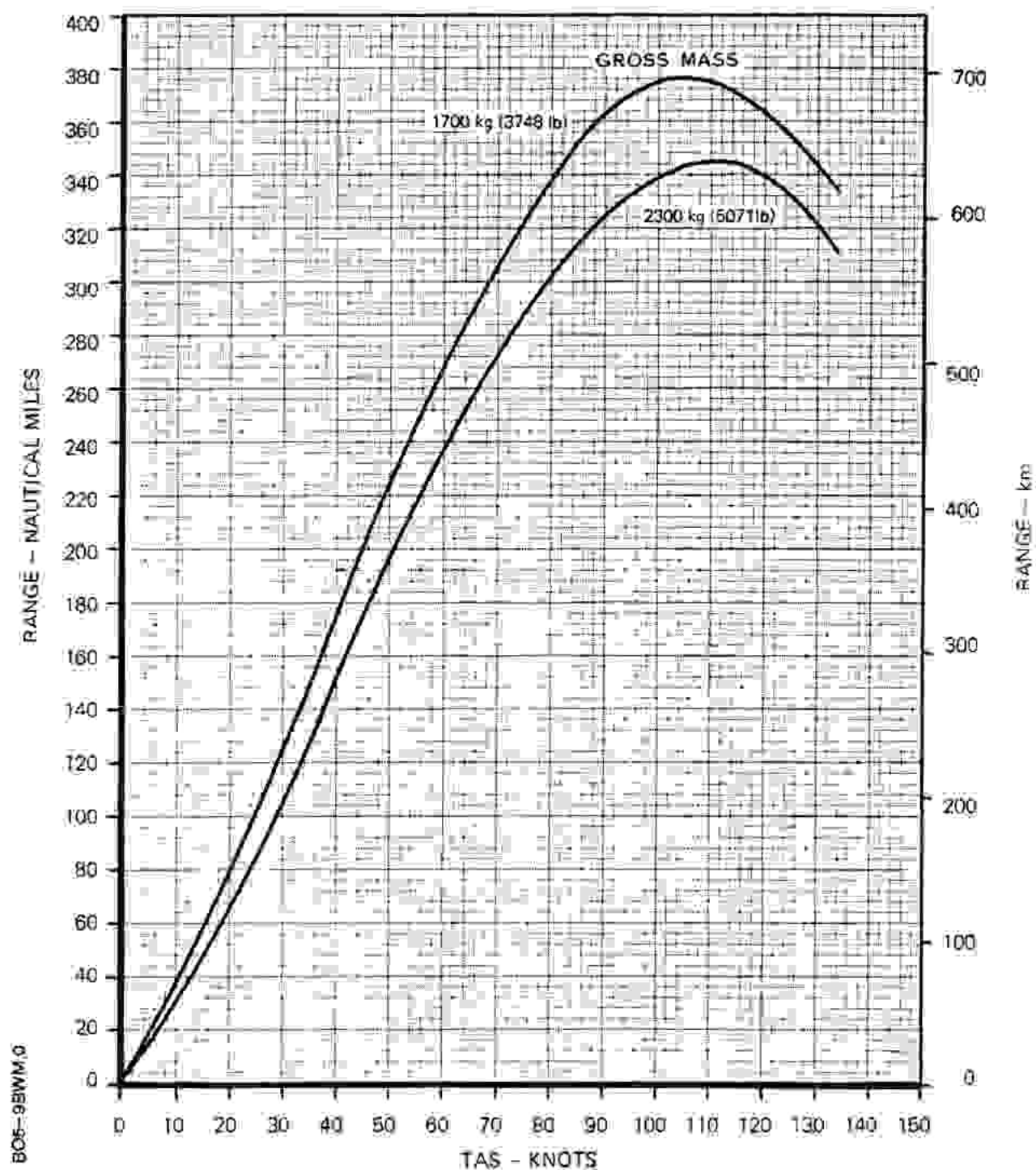


Fig. 9-11 Maximum Range - ISA 5000 ft

MANUFACTURER'S DATA

Rev. 0

9.4 MAXIMUM ENDURANCE

The maximum endurance diagrams (see Fig. 9-12 and 9-13) are based on the following conditions:

- International Standard Atmosphere (ISA)
- Usable fuel - 456 kg (no reserve)
- Level flight (no climb, descent, or other maneuver)
- Engine anti-icing - OFF

NOTE With anti-icing ON, the maximum endurance will be reduced.

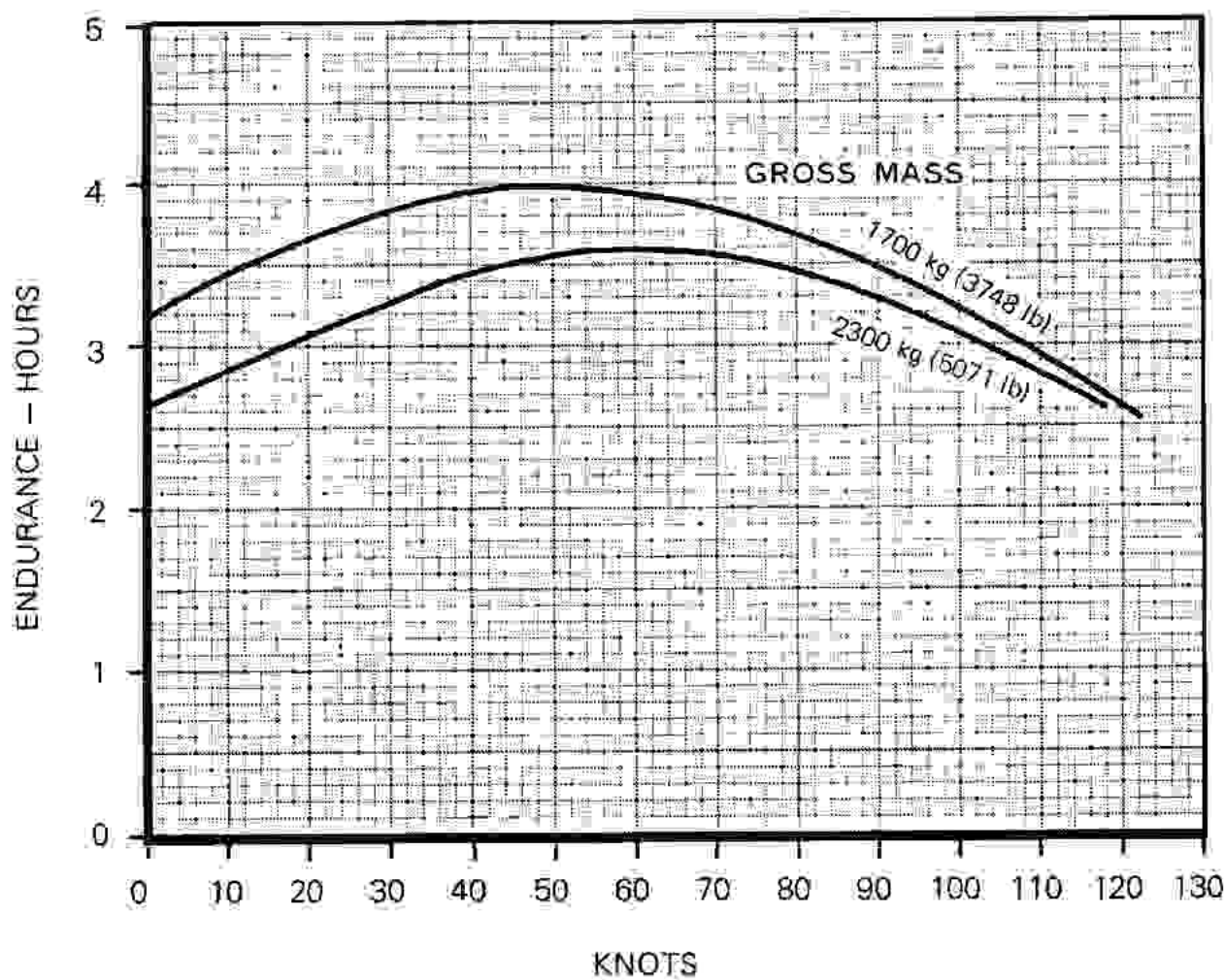
MAXIMUM ENDURANCE

PRESSURE ALTITUDE - ISA SEA LEVEL

2 X ALLISON 250-C20B

FUEL - 456 KG, NO RESERVE

ANTI-ICING - OFF



BO5-9BYM.0

Fig. 9-12 Maximum Endurance - ISA Sea Level

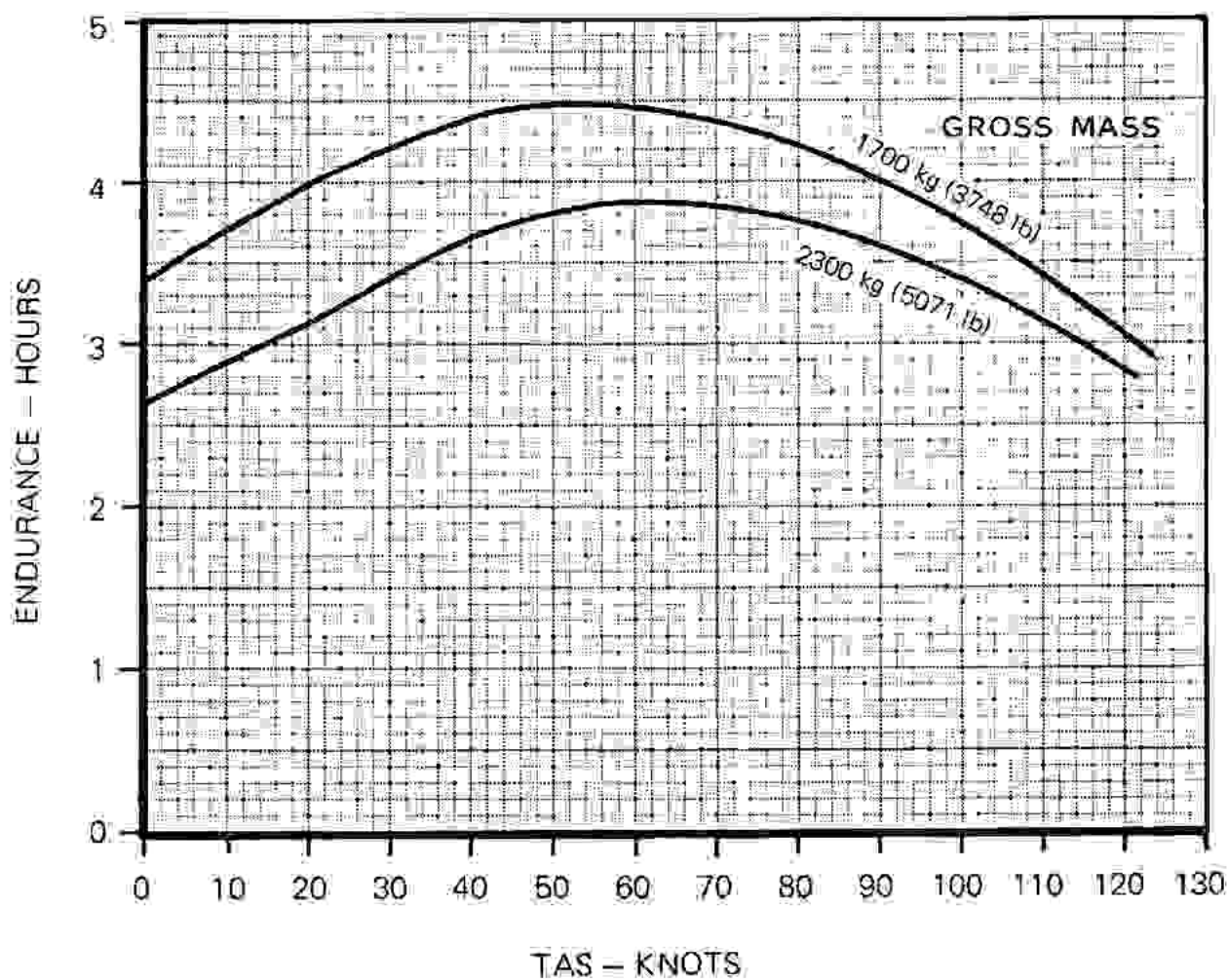
MAXIMUM ENDURANCE

PRESSURE ALTITUDE - ISA 5000 FT

2 X ALLISON 250-C20B

FUEL - 456 KG, NO RESERVE

ANTI-ICING - OFF



BO5-9BAP.0

Fig. 9-13 Maximum Endurance - ISA 5000 ft

MANUFACTURER'S DATA

9.5 MAXIMUM CRUISING SPEED

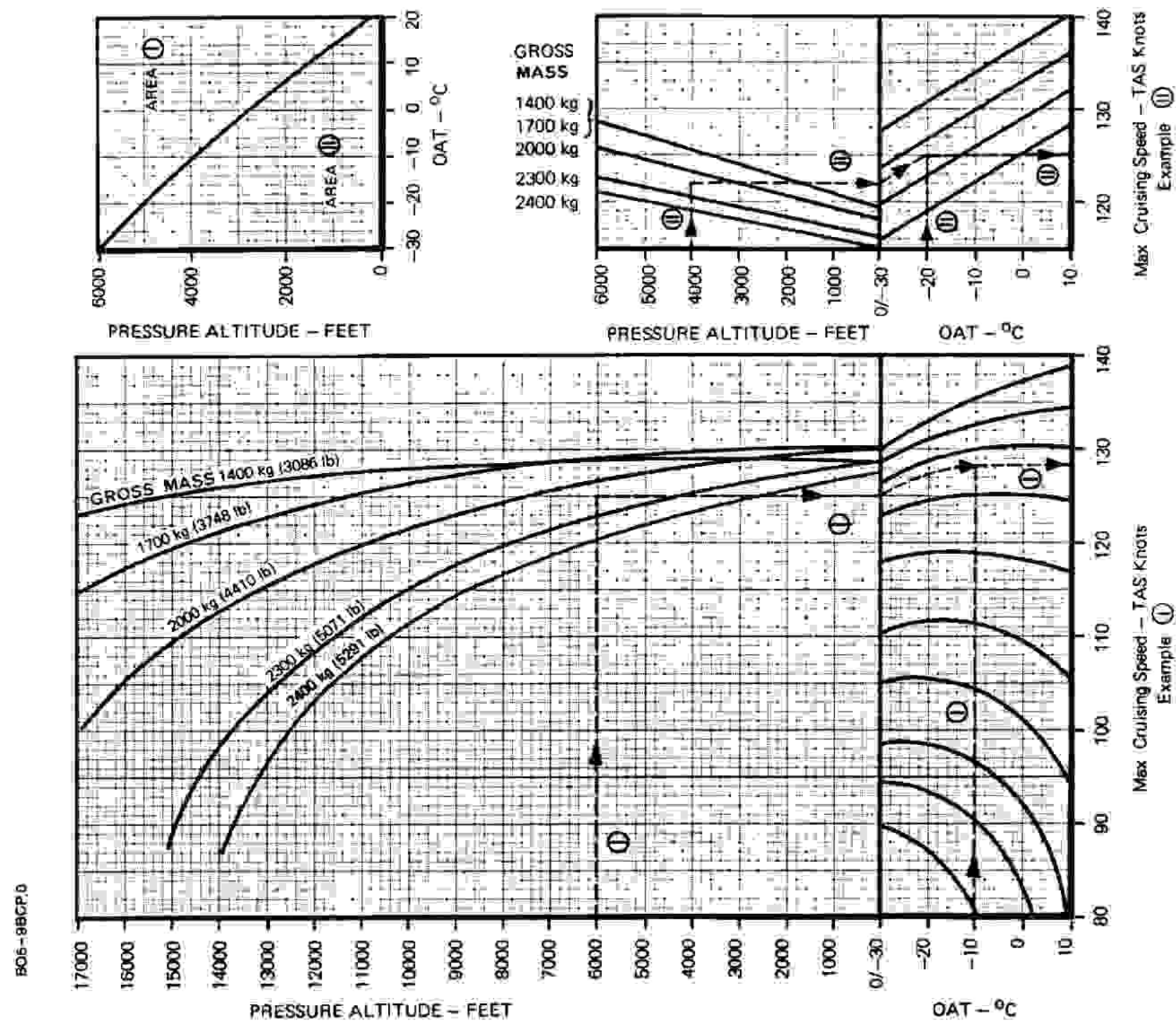
The maximum cruising speed diagrams (see Fig. 9-14 and 9-15) are based on both engines operating at maximum continuous power.

MAXIMUM CRUISING SPEED

2 X ALLISON 250-C20B

AEO - MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING - ON



NOTE: EXAMPLE I (PRESSURE ALTITUDE 6000 FT, GROSS MASS 2150 KG, OAT -10°C, MAX CRUISING SPEED = 128 KTAS) IS APPLICABLE WHEN INTERSECTION OF PRESSURE ALTITUDE AND OAT IN THE DIAGRAM ABOVE RIGHT IS WITHIN AREA I.

EXAMPLE II (PRESSURE ALTITUDE 4000 FT, GROSS MASS 2150 KG, OAT -20°C, MAX CRUISING SPEED = 125 KTAS) IS APPLICABLE WHEN INTERSECTION OF PRESSURE ALTITUDE AND OAT IN THE DIAGRAM ABOVE RIGHT IS WITHIN AREA II.

VNE-TABLE VALUES WILL ALWAYS LIMIT THE MAX CRUISE SPEED VALUES FOUND ABOVE

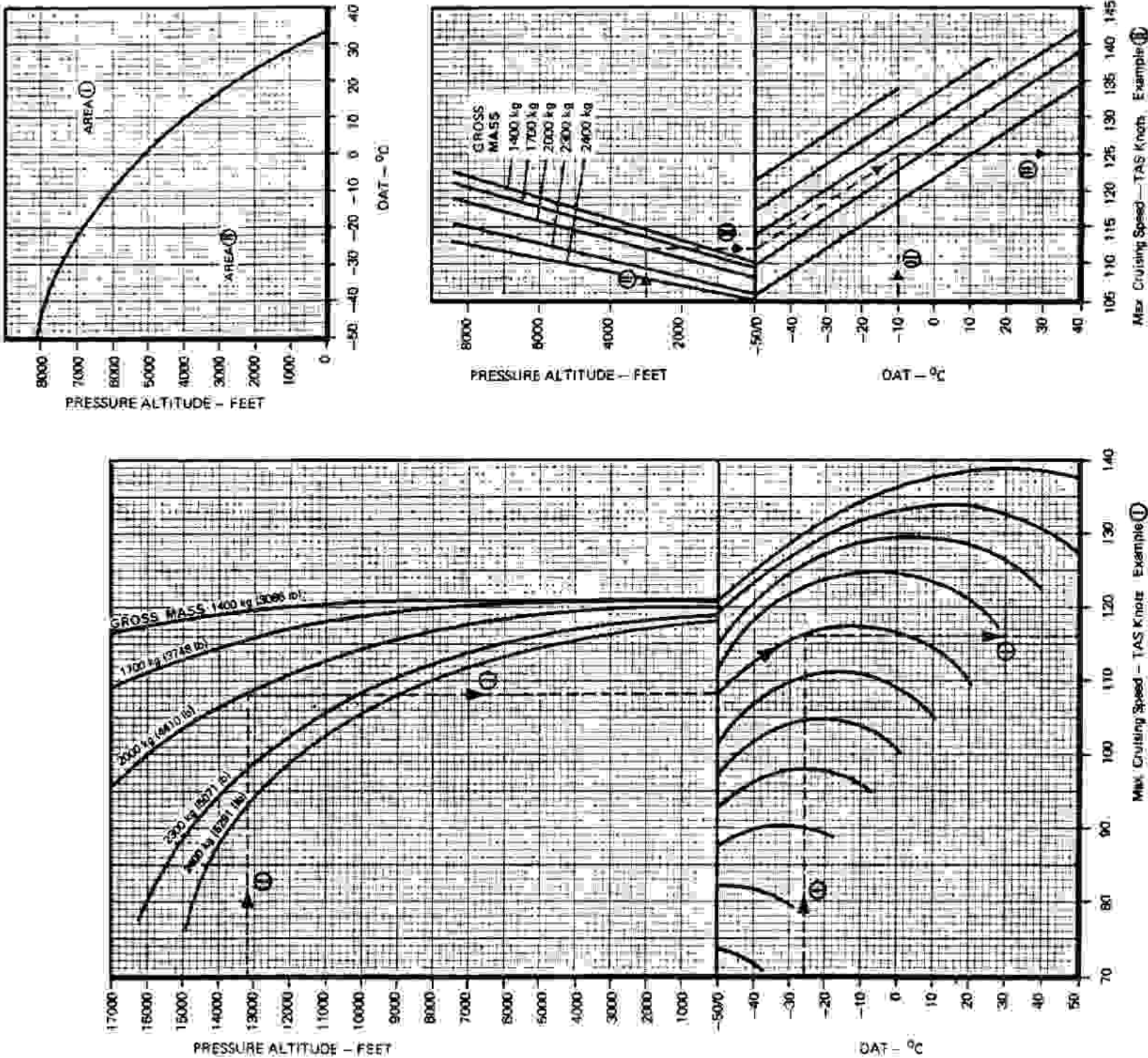
Fig. 9-14 Maximum Cruising Speed - Anti-Icing ON

MAXIMUM CRUISING SPEED

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF



NOTE: EXAMPLE I (PRESSURE ALTITUDE 13200 FT, GROSS MASS 2000 KG, OAT -26°C, MAX CRUISING SPEED = 116 KTAS) IS APPLICABLE WHEN INTERSECTION OF PRESSURE ALTITUDE AND OAT IN THE DIAGRAM ABOVE RIGHT IS WITHIN AREA I

EXAMPLE II (PRESSURE ALTITUDE 3000 FT, GROSS MASS 2000KG, OAT -10°C, MAX CRUISING SPEED = 125 KTAS) IS APPLICABLE WHEN INTERSECTION OF PRESSURE ALTITUDE AND OAT IN THE DIAGRAM ABOVE RIGHT IS WITHIN AREA II

VNE-TABLE VALUES WILL ALWAYS LIMIT THE MAX CRUISE SPEED VALUES FOUND ABOVE

Fig. 9-15 Maximum Cruising Speed - Anti-icing OFF

9.6 OEI MASS/ALTITUDE/TEMPERATURE

The one-engine inoperative mass/altitude/temperature diagrams (see Fig. 9-16 and 9-17) are based on having a climb reserve of 50 and 150 fpm at a height of 1000 ft above the takeoff field-elevation and at the best rate-of-climb speed of 60 KIAS.

OEI MASS/ALTITUDE/TEMPERATURE

50-FPM CLIMB RESERVE

1 X ALLISON 250-C20B

OEI - EMERGENCY POWER (810 °C TOT, 95% TORQUE)

ANTI-ICING - OFF

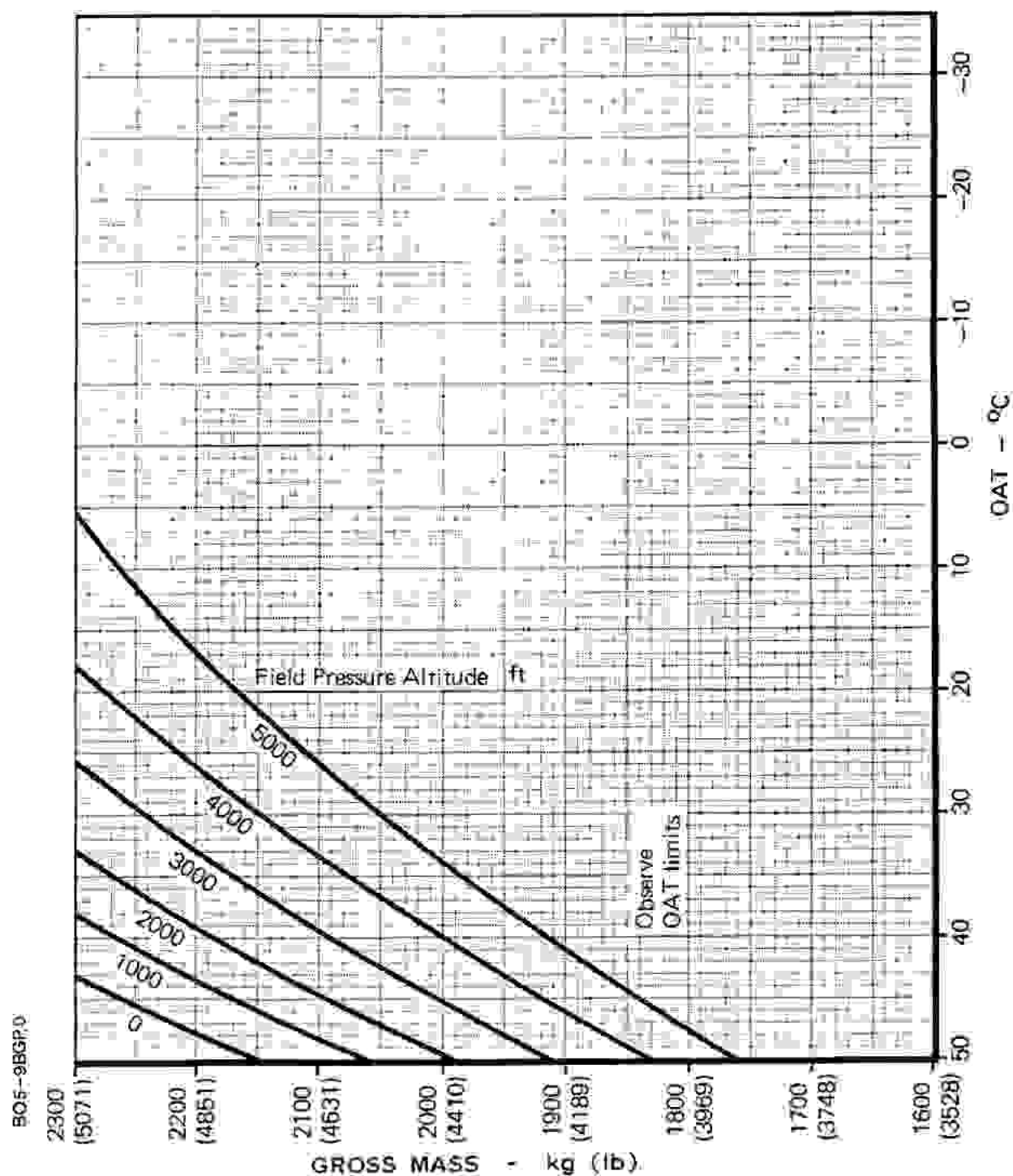


Fig. 9-16 OEI Mass/Altitude/Temperature - 50-fpm Climb Reserve

MANUFACTURER'S DATA

OEI MASS/ALTITUDE/TEMPERATURE

150-FPM CLIMB RESERVE

1 X ALLISON 250-C20B

OEI - EMERGENCY POWER (810 °C TOT, 95% TORQUE)

ANTI-ICING - OFF

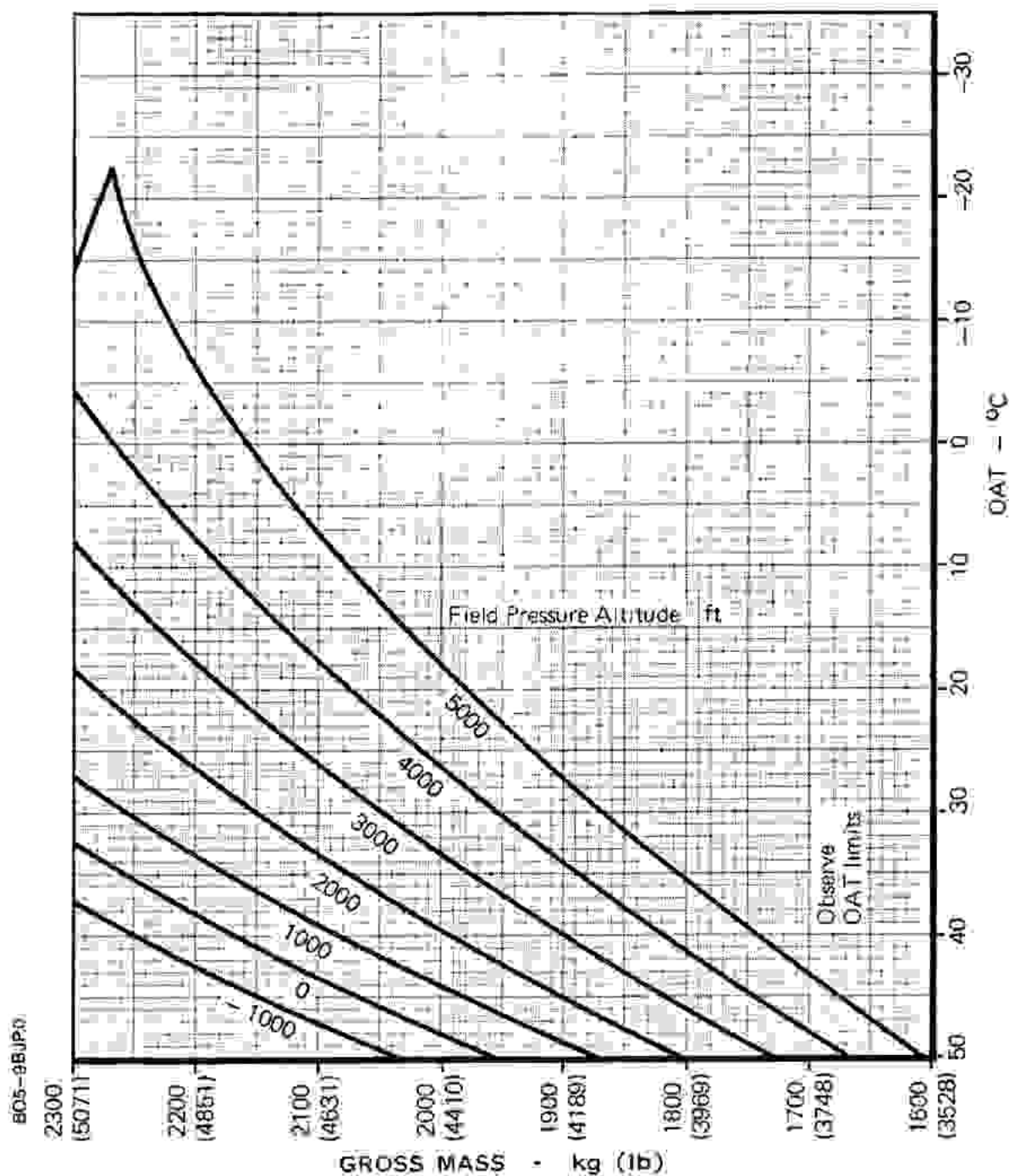


Fig. 9-17 OEI Mass/Altitude/Temperature - 150-fpm Climb Reserve

9.7 GUIDELINES FOR LOW NOISE OPERATIONS

The following guidelines are used to operate the BK 117 BO 105 helicopters in noise sensitive areas. These guidelines are recommendations only. The flight procedures remain under the pilot's responsibility, according to flight manual limitations and local regulation restrictions.

9.7.1 General

The helicopter is noisiest while hovering, immediately after take-off and just after flaring immediately before landing. Therefore minimize the time spent in these positions.

Adopt a flight path as far as possible from sensitive areas.

For flights over sensitive areas prefer a flight path along the noisiest route (motorway, railway, etc.).

Noise emission is lower on the left side of the helicopter, to the sides of the flight path than directly underneath and upwind than downwind of the helicopter.

9.7.2 Operating in sensitive areas

Take-off and climb

Climb to cruise altitude at V_Y

Overflights

When crossing noise-sensitive areas maximum $V_{IAS} = 125$ kts.

Plan routes to keep noise-sensitive areas on the left side of the helicopter.

Where possible fly at least 1000 ft AGL.

Turns

Avoid quick and repetitive pedal movements and prefer right-stop-turn to left-stop-turn.

Descent, approach and landing

Near noise sensitive areas use a steep angle of descent (approx. 12-15°).

Land quickly after flaring and do not hover.

Plan the approach and landing to keep noise-sensitive areas forward and to the left side of the helicopter.

Avoid descending directly over noise-sensitive areas.

9.7.3 Maneuver near the ground (hovering)

Avoid unnecessary hovering.

Avoid quick and repetitive pedals movement.